

BINARY ARITHMETIC PRACTICE SHEET

NIMCET TARGET PAPER • ADVANCED / HARD LEVEL

ResourceOS

1. In IEEE 754 single-precision 32-bit floating-point format, what is the hex representation for the decimal number -12.625 ?

- (A) C14A0000 (B) 414A0000
(C) C12A0000 (D) C0CA0000

2. Using Booth's Multiplication Algorithm for signed integers, what operation is performed when the current multiplier bit Q_0 and previous implicit bit Q_{-1} are 10?

- (A) Subtract Multiplicand from Accumulator ($AC = AC - M$) (B) Add Multiplicand to Accumulator ($AC = AC + M$)
(C) Arithmetic Shift Right only (D) Logical Shift Left

3. What is the biased exponent value stored in the 8-bit exponent field of an IEEE 754 single-precision float for decimal 0.15625 ?

- (A) 124 (01111100) (B) 127 (01111111)
(C) 130 (10000010) (D) 121 (01111001)

4. In Non-Restoring Division algorithm, if the updated partial remainder after subtraction becomes negative, the next step involves:

- (A) Set quotient bit to 0 and ADD divisor in the next step (B) Immediately add divisor back before shifting
(C) Set quotient bit to 1 and SUBTRACT divisor (D) Abort division due to overflow

5. How many maximum Booth's recording transitions (add/subtract operations) occur when multiplying by the 8-bit multiplier 01010101?

- (A) 8 operations (B) 4 operations
(C) 2 operations (D) 0 operations

6. Convert the binary string 101101_2 to its corresponding Gray Code.

- (A) 111011 (B) 110111
(C) 100110 (D) 111100

7. What is the result of adding two 8-bit signed 2's complement numbers 10000000 (-128) and 10000000 (-128)?

- (A) 00000000 with Overflow = 1 and Carry = 1 (B) 10000000 with Overflow = 0 and Carry = 1
(C) 11111111 with Overflow = 1 and Carry = 0 (D) 00000000 with Overflow = 0 and Carry = 0

8. Convert Gray code sequence 110101_{Gray} back into Binary code.

- (A) 100110 (B) 101001
(C) 111001 (D) 100011

9. In an n -bit Carry Lookahead Adder (CLA), the Carry Generate function G_i and Carry Propagate function P_i for stage i are defined as:

- (A) $G_i = A_i \cdot B_i$, $P_i = A_i \oplus B_i$ (B) $G_i = A_i + B_i$, $P_i = A_i \cdot B_i$
(C) $G_i = A_i \oplus B_i$, $P_i = A_i \cdot B_i$ (D) $G_i = A_i \cdot B_i$, $P_i = A_i + B_i$

10. In 32-bit IEEE 754 standard, what decimal value does the hex word 0x00000000 represent versus 0x80000000?

- (A) +0 and -0 respectively
- (B) +0 and NaN respectively
- (C) +1 and -1 respectively
- (D) Underflow and Overflow

11. If two 8-bit IEEE 754-like floating point numbers have 1 sign bit, 3 exponent bits (bias = 3), and 4 mantissa bits, what is decimal 3.5 in this system?

- (A) 0 100 1100
- (B) 0 101 0110
- (C) 0 011 1100
- (D) 0 100 0110

12. What is the maximum relative precision error (machine epsilon) for a floating point system with a t -bit normalized fraction mantissa?

- (A) 2^{-t}
- (B) $2^{-(t+1)}$
- (C) 2^{1-t}
- (D) 2^t

13. Perform bitwise binary multiplication of 11.011_2 by 1.01_2 .

- (A) 100.00111
- (B) 100.01111
- (C) 10.11101
- (D) 101.00111

14. Which condition in IEEE 754 single-precision float represents +Infinity?

- (A) Exponent = 255 (all 1s), Mantissa = 0, Sign = 0
- (B) Exponent = 255 (all 1s), Mantissa $\neq$ 0, Sign = 0
- (C) Exponent = 0, Mantissa = 0, Sign = 0
- (D) Exponent = 127, Mantissa = all 1s, Sign = 0

15. What is the 8-bit 2's complement result of evaluation: $(\sim 0 \times 35) + 1$ where $\sim$ represents bitwise NOT?

- (A) 11001011
- (B) 11001010
- (C) 00110101
- (D) 11010101

16. Compute fractional division $0.11_2 / 0.1_2$ in binary representation.

- (A) 1.1
- (B) 0.11
- (C) 1.01
- (D) 0.01

17. In a 4-bit synchronous binary counter using 2's complement logic, what is the sequence state immediately following 0111 (+7) during increment?

- (A) 1000 (-8)
- (B) 1001 (-7)
- (C) 0000 (0)
- (D) 1111 (-1)

18. Which of the following numbers CANNOT be represented exactly as a finite binary floating-point expansion?

- (A) 0.2
- (B) 0.625
- (C) 0.875
- (D) 0.0625

19. When subtracting $B = 10110_2$ from $A = 01101_2$ using 2's complement addition, what is the status of Carry Out (C) and Overflow (V)?

- (A) $C = 0, V = 0$ (B) $C = 1, V = 0$
(C) $C = 0, V = 1$ (D) $C = 1, V = 1$

20. In a 16-bit computer system using Excess-127 representation for exponents, what exponent value corresponds to a stored field of 10000011?

- (A) +4 (B) +131
(C) -4 (D) +3

21. Multiply signed 2's complement numbers 1110 (-2) and 1101 (-3) using 4-bit signed binary arithmetic.

- (A) 00000110 (+6) (B) 11111010 (-6)
(C) 00001110 (+14) (D) 11100110 (-26)

22. A denormalized (or subnormal) number in IEEE 754 single-precision float has an exponent field equal to:

- (A) 00000000 with non-zero mantissa (B) 11111111 with zero mantissa
(C) 01111111 with non-zero mantissa (D) 00000001 with zero mantissa

23. Calculate the 8-bit bitwise expression: $(0x8A \text{ XOR } 0x55) \text{ AND } (\text{NOT } 0x0F)$.

- (A) 11010000 (0xD0) (B) 11011111 (0xDF)
(C) 10100000 (0xA0) (D) 00001111 (0x0F)

24. In a 6-bit 2's complement code system, what is the decimal equivalent of the largest negative number that can be stored?

- (A) -32 (B) -64
(C) -31 (D) -63

25. In carry-save addition (CSA) tree used in fast array multipliers, three n -bit inputs are reduced to:

- (A) Two numbers (Sum vector and Carry vector) without carry propagation (B) One single combined sum word in 1 clock cycle
(C) Three modified partial products (D) An n -bit ripple carry output

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(A)	6	(A)	11	(A)	16	(A)	21	(A)
2	(A)	7	(A)	12	(A)	17	(A)	22	(A)
3	(A)	8	(A)	13	(A)	18	(A)	23	(A)
4	(A)	9	(A)	14	(A)	19	(A)	24	(A)
5	(A)	10	(A)	15	(A)	20	(A)	25	(A)